RUIQI HE

CONTACT

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EDUCATION

The Ohio State University|Columbus, OH — **Expected Graduation: May 2028**
Bachelor of Science, Data Analytics | Specialization: Biomedical and Public Health Analytics
Minor: Business|ASC Honors College | Dean's List

SKILLS

- Programming Languages: Python, SQL, R, Java, C++, TypeScript/JavaScript
- Machine Learning & AI: PyTorch, scikit-learn, Pandas, prompt engineering, LLMs (Gemini, Qwen, IBM watsonx), NVIDIA RAPIDS cuDF
- Data Engineering & Analytics: Databricks, geospatial analysis, large-scale preprocessing, Monte Carlo / VaR–ES
- Software & Tools: FastAPI, React, Node.js/Express, Docker, Git/GitHub Actions, NVIDIA DGX
- Spoken Languages: English (Native), Chinese (Native), Spanish (Basic)

EXPERIENCE

Undergraduate Research Intern – AIMed Lab | The Ohio State University | Department of Biomedical — ***Summer 2026***
Mentor: Dr. Ping Zhang

- Competing in the 2026 George B. Moody PhysioNet Challenge to develop predictive models that forecast cognitive impairment diagnoses from polysomnography (sleep study) data
- Engineering ML pipelines to analyze multi-modal physiological signals including EEG, EOG, EMG, ECG, respiratory effort, and airflow to identify patients at risk for mild cognitive impairment, Alzheimer's disease, and dementia
- Processing large-scale PSG datasets (1.2TB training set across 6,600+ recordings) with heterogeneous channel availability, variable sampling rates, and real-world noise
- Optimizing age-conditioned AUROC scoring metrics to predict diagnoses occurring 1-6 years post-sleep study

PROJECTS

ASA DataFest @ OSU — 1st Place, Best Visualization | Spring 2026

- **Analyzed 7.7M+ healthcare encounters across 948K+ patients from four years of Stormont Vail Health data.**
- **Built a preprocessing pipeline with NVIDIA RAPIDS cuDF on DGX Spark, accelerating cleaning, joins, and transformation of large healthcare data.**
- **Integrated encounters, diagnoses, providers, departments, SDOH, and census-tract data into a unified analytical dataset.**
- **Helped build an interactive geospatial dashboard with regional selection, diagnosis filters, patient distribution, and access-trend analysis.**
- **Translated complex patient-journey patterns into intuitive visualizations, earning 1st Place for Best Visualization.**

CLARA (Clairvoyant Loss Avoidance & Risk Advisor) | Hackathon Win — ***Spring 2026***

- Built a full-stack enterprise risk platform combining quantitative risk modeling with AI analysis of SEC filings.
- Implemented VaR, Expected Shortfall, Monte Carlo simulation, stress testing, and sensitivity analysis across multiple distributions.
- Automated SEC EDGAR 10-K Item 1A extraction and COSO ERM risk classification using IBM watsonx.
- Developed interactive portfolio dashboards with likelihood-impact heat maps, sensitivity views, and multi-confidence risk analysis.
- Integrated Gemini and IBM watsonx for conversational portfolio-risk analysis and company-disclosure insights.

Research Society — Qwen Cloud Hackathon — ***Summer 2026***

- Built a five-agent Qwen research-review system for methodology, novelty, evidence, literature context, and research-gap analysis.
- Designed a real-time debate framework with rebuttals, consensus voting, and a dissent ledger to preserve unresolved disagreements.
- Developed a FastAPI + React + WebSocket application for live agent reasoning, debate events, and review summaries.
- Added paper ingestion from PDF, arXiv, DOI, URL, OpenAlex, PubMed, Crossref, and natural-language search.
- Containerized with Docker and deployed on Alibaba Cloud ECS, using Qwen through DashScope for inference.

Aegis — Security Automation Pipeline | GitLab AI Hackathon — ***Spring 2026***

- Built a multi-agent security pipeline that detects, triages, fixes, validates, and commits vulnerabilities in GitLab CI/CD.
- Integrated SAST, DAST, SCA, secrets detection, and IaC scanning into a unified automated remediation workflow.
- Designed Triage → Fix → Validate agent pipelines with scoped permissions and minimal unified-diff generation.
- Restricted fix agents to read-only access while validation agents handled repository changes and commits.
- Added guardrails that auto-commit safe fixes while escalating destructive IaC changes for human review.

Universe-616 Multi-Agent AI System | Personal Project — ***2026***

- Built a hierarchical multi-agent AI system with specialized agents, task routing, permission gates, and dynamic delegation.
- Integrated vLLM on NVIDIA DGX Spark for private local inference, long-context workflows, and concurrent agent execution.

ACTIVITIES

- BDAA(Big Data & Analytics Association): Autumn 2025 - Present
  - Foundation of AI TA: Spring 2026 - Present
- Formula Buckeye: Autumn 2025 - Present
- Academic Team: Autumn 2022 – Spring 2024 (PA State Academic Team Champion)
- Mock Trial: Autumn 2022 – Spring 2024